

COIMBRA, Portugal

WMO: 085490

Lat: 40.2135N

Lon: 8.4552W

Elev: 115

StdP: 14.64

Time Zone: 0.00 (EUW)

Period: 86-96

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 36.2	(c) 38.3	(d) 26.4	(e) 20.4	(f) 45.1	(g) 29.1	(h) 23.2	(i) 45.0	(j) 26.9	(k) 53.2	(l) 23.4	(m) 54.2	(n) 3.8	(o) 180	(p) 0.394

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 21.9	(c) 93.3	(d) 70.2	(e) 88.7	(f) 69.1	(g) 84.8	(h) 68.1	(i) 72.3	(j) 88.1	(k) 70.6	(l) 85.2	(m) 69.2	(n) 82.3	(o) 6.3	(p) 310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 66.8	(b) 99.1	(c) 78.1	(d) 65.6	(e) 94.9	(f) 75.5	(g) 64.6	(h) 91.6	(i) 73.6	(j) 35.9	(k) 88.3	(l) 34.4	(m) 85.4	(n) 33.2	(o) 82.8	(p) 82.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 21.7	(b) 16.8	(c) 14.1	(e) 32.5	(f) 101.6	(g) 2.1	(h) 2.4	(i) 31.0	(j) 103.3	(k) 29.8	(l) 104.7	(m) 28.6	(n) 106.1	(o) 27.1	(p) 107.9		
(a) 30.1	(b) 76.5	(c) 2.3	(e) 30.1	(f) 76.5	(g) 2.3	(h) 2.9	(i) 28.4	(j) 78.6	(k) 27.1	(l) 80.2	(m) 25.8	(n) 81.9	(o) 24.2	(p) 83.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	60.9	50.1	52.0	56.2	56.5	63.0	67.3	72.4	71.8	68.7	62.4	56.5	52.9	(6)
(7)		DBStd	9.12	4.11	4.72	5.08	5.69	5.67	5.14	5.48	4.80	5.38	5.42	5.31	4.62	(7)
(8)		HDD50	131	49	31	9	7	0	0	0	0	0	0	10	25	(8)
(9)		HDD65	2284	463	364	274	262	107	31	3	4	23	118	258	377	(9)
(10)		CDD50	4100	52	86	202	202	403	520	694	676	561	385	205	114	(10)
(11)		CDD65	778	0	0	3	7	45	101	232	215	134	38	3	0	(11)
(12)		CDH74	5843	0	0	22	86	394	761	1983	1571	849	176	1	0	(12)
(13)		CDH80	2154	0	0	1	15	120	249	871	578	287	33	0	0	(13)
(14)	Wind	WSAvg	5.2	5.7	5.3	5.5	5.1	5.0	4.7	4.7	4.5	4.5	5.4	5.3	6.2	(14)
(15)	Precipitation	PrecAvg	36.3	5.2	4.0	3.1	3.3	3.0	1.2	0.4	0.6	2.0	4.6	4.7	4.5	(15)
(16)		PrecMax	50.0	15.2	11.5	11.4	9.4	7.4	3.6	2.0	2.1	5.4	10.6	16.4	14.1	(16)
(17)		PrecMin	21.6	0.4	0.6	0.0	1.1	0.0	0.0	0.0	0.0	0.0	0.7	0.4	0.3	(17)
(18)		PrecStd	8.3	3.7	2.7	2.7	2.0	1.9	0.9	0.5	0.6	1.5	2.6	3.4	3.1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	63.0	69.3	77.1	82.0	89.1	92.1	98.6	97.9	94.8	84.4	72.4	65.4	(19)
(20)		MCWB	54.2	55.7	60.2	62.4	65.0	69.4	71.7	71.4	69.9	65.5	61.8	58.7	(20)	
(21)		2%	DB	60.4	64.4	73.0	75.9	83.4	86.4	94.3	91.0	87.6	78.7	69.6	62.8	(21)
(22)		MCWB	54.0	53.8	59.1	60.0	64.3	68.8	71.1	69.8	68.3	63.5	60.8	57.6	(22)	
(23)		5%	DB	58.4	61.7	69.1	70.9	78.0	82.2	89.3	86.2	82.4	74.8	66.7	61.2	(23)
(24)		MCWB	53.5	53.4	57.2	58.5	62.5	67.3	70.1	68.8	66.0	62.6	59.7	57.3	(24)	
(25)		10%	DB	56.8	59.3	65.6	66.8	73.5	77.9	84.3	81.8	78.4	71.5	64.6	59.9	(25)
(26)		MCWB	52.5	53.1	55.4	56.4	61.4	65.3	68.9	67.7	64.9	61.5	58.5	56.0	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.5	59.7	62.8	64.5	67.4	73.0	75.1	73.4	71.5	67.1	65.5	61.6	(27)
(28)		MDCB	59.6	63.0	72.9	78.1	80.6	86.7	94.4	90.9	87.8	78.0	68.2	62.7	(28)	
(29)		2%	WB	56.7	57.5	59.8	61.6	65.4	70.2	72.7	71.6	69.5	65.5	62.8	60.2	(29)
(30)		MDCB	58.3	60.6	70.3	72.9	79.5	84.0	89.1	86.7	81.8	74.1	66.4	61.6	(30)	
(31)		5%	WB	55.2	55.7	58.1	59.4	63.7	68.1	71.0	70.0	68.0	64.1	60.9	58.6	(31)
(32)		MDCB	57.1	58.6	67.4	68.2	76.2	79.8	86.2	84.0	79.0	70.7	65.3	60.5	(32)	
(33)		10%	WB	53.6	54.2	56.5	57.5	61.8	66.3	69.2	68.5	66.8	62.8	59.6	56.6	(33)
(34)		MDCB	55.7	57.5	63.7	64.7	72.0	76.2	83.3	80.5	75.9	68.5	63.6	58.9	(34)	
(35)	Mean Daily Temperature Range	MDBR	12.5	13.5	17.0	16.6	17.8	18.9	21.9	21.3	19.2	14.3	12.7	10.5	(35)	
(36)		5% DB	MCDBR	14.5	17.0	23.1	23.8	25.8	26.4	30.2	28.3	26.8	20.6	15.3	11.2	(36)
(37)		MCWBR	9.4	9.5	12.0	11.6	12.0	12.1	12.2	11.6	11.3	9.1	8.6	7.5	(37)	
(38)		5% WB	MCDBR	9.6	12.1	20.3	21.7	22.5	23.6	26.8	24.6	21.7	16.7	11.3	7.6	(38)
(39)		MCWBR	7.6	8.3	11.2	11.0	11.2	11.4	12.1	10.7	9.5	8.2	7.4	6.3	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.330	0.343	0.377	0.377	0.384	0.389	0.377	0.394	0.382	0.369	0.340	0.327	(40)	
(41)		taud	2.503	2.457	2.367	2.378	2.384	2.389	2.405	2.359	2.417	2.471	2.506	2.510	(41)	
(42)		Ebn at Noon	264	277	277	285	284	281	284	276	272	264	258	255	(42)	
(43)		Edh at Noon	24	29	35	37	38	38	37	37	33	28	24	23	(43)	
(44)	All-Sky Solar Radiation	RadAvg	640	958	1292	1672	2046	2251	2321	2095	1626	1066	714	580	(44)	
(45)		RadStd	102	132	175	196	191	144	99	88	125	116	117	83	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (20 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+103

Nomenclature: See separate page